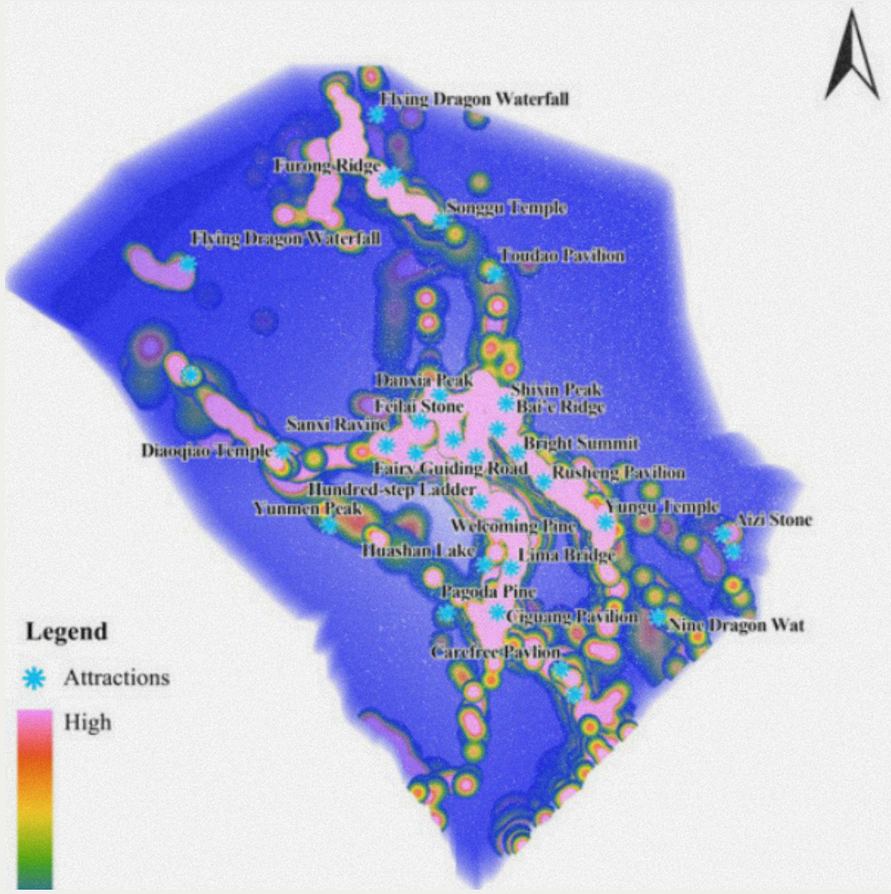
MAP FRESHNESS AUDIT

May 2026
HERE HACKHATON



THE REALITY GAP

Is Your Map Lying to You??

THE "GHOST SIGN" RISK

MAPS ARE OFTEN OUTDATED. SPEED SIGNS ARE REMOVED PHYSICALLY, BUT STAY IN THE DATABASE FOR MONTHS.
THIS CREATES CRITICAL FRICTION FOR INTELLIGENT SPEED ASSISTANCE (ISA) SYSTEMS, LEADING TO DANGEROUS PHANTOM BRAKING AND LOSS OF USER TRUST.

PROACTIVE AUDITING

INSTEAD OF WAITING FOR MANUAL REPORTS, WE USE A CONTINUOUS "DOUBLE-CHECK" PIPELINE: VALIDATING WHAT THE MAP SAYS AGAINST WHAT DRIVERS DO.



OUR MISSION

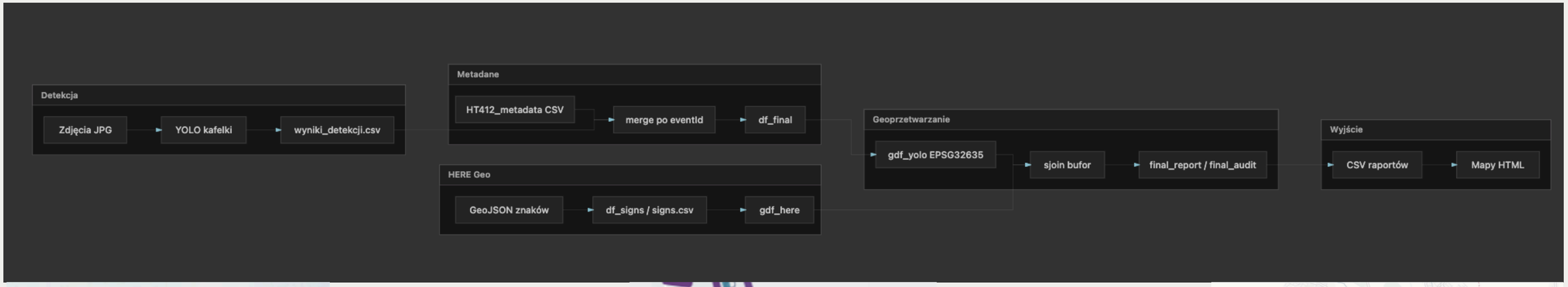
Audit Accuracy

Develop an automated pipeline to identify "Map Lies" by comparing physical detections with real-world driver behavior.

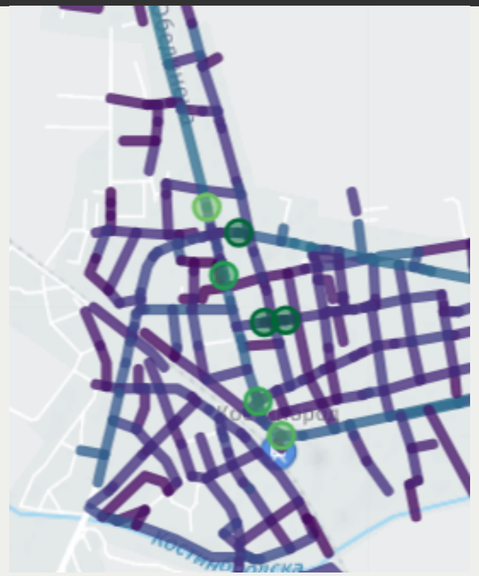
Safety First

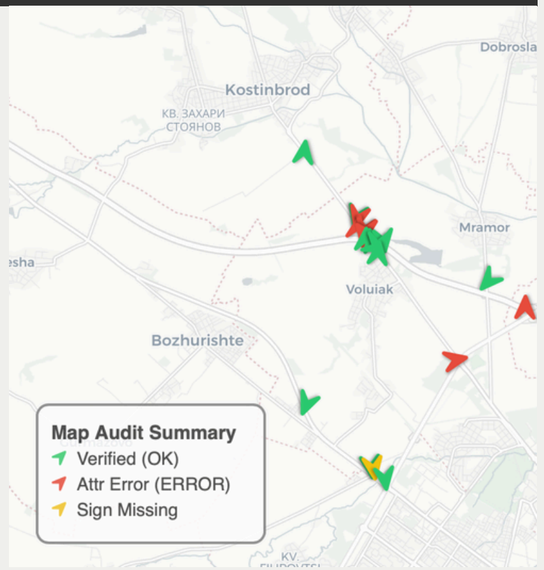
Ensure that autonomous vehicles and speed assistants operate on 100% verified signage, reducing liability and accidents.

THE MULTI-MODAL PIPELINE









PRECISION GIS FILTERING

12m Precision Buffer
Standard GPS noise can merge data from parallel roads. We implement a 12-meter Lane Buffer in UTM Projection to isolate the exact traffic flow facing the sign.

Directional Gate
We filter vehicles by Heading vs Bearing (tolerance +/- 20°). This ensures we ignore cars on the opposite side of the road.

Route-scoped audit
We only evaluate signs along the driven footprint, so cost scales with collected mileage, not the whole map.

Tiled inference + NMS + empty-tile skip
Handles large photos reliably, removes duplicate boxes, and cuts wasted compute on textureless tiles.

Aggregation per HERE sign ID
Many frames collapse to one stable label + confidence summary per map object — less noise than trusting a single image.

DATA PROCESSING

Problem of few samples
For some signs, only a few samples existed, which could impact the reliability of our statistics.To ensure our conclusions are strictly evidence-based and statistically sound, we rejected data with an insufficient sample size.

Night drivers
In the night, drivers often drive much faster, because of the lack of car traffic. We rejected those samples in order to get more reliable statistic.

85th Percentile Strategy
Instead of relying on average speeds, we focus on the 85th percentile to represent "free-flow" traffic. In traffic engineering, this value shows the speed that 85% of drivers naturally choose. If 85% of drivers are significantly exceeding the limit, it is a strong indicator that the map data is outdated or the sign is missing.

Confidence Score Logic
The Confidence Score is our final metric for sign reliability, calculated by comparing the 85th percentile speed to the official speed limit. A high score (near 100%) confirms that drivers are following the map data, meaning the sign is likely correct. A low score indicates a major mismatch, suggesting the sign is either missing or incorrectly recorded in the database.

VISUAL EXTRACTION

SAHI + YOLO Inference

We process high-resolution street-level imagery using **Slicing Aided Hyper Inference** (SAHI).

- ✓ Detects small objects in complex urban grids
- ✓ Aggregates sign data within 45m search buffers
- ✓ Merges detections with HERE Map Metadata



EXTENDED DESCRIPTION

STEP 1: COMPUTER VISION & SAHI INFERENCE

WE PROCESS HIGH-RESOLUTION STREET-LEVEL IMAGERY USING SAHI (SLICING AIDED HYPER INFERENCE)COMBINED WITH YOLO. THIS ALLOWS US TO DETECT SMALL OBJECTS (LIKE SPEED LIMIT SIGNS) WITH EXTREMEPRECISION, EVEN IN COMPLEX URBAN ENVIRONMENTS.

STEP 2: SPATIAL ASSOCIATION & DATA AGGREGATION

WE MERGE DETECTED SIGNS WITH EXISTING HERE MAP METADATA (SIGNS_SPDLMT).

A 45M SEARCH BUFFER IS APPLIED FOR INITIAL SPATIAL ASSOCIATION TO ENSURE WE CAPTURE ALL NEARBYIMAGERY RELEVANT TO THE EXISTING MAP NODES.

STEP 3: BEHAVIORAL VALIDATION (THE PROBE LAYER)

WE INTEGRATE RAW HERE PROBE DATA AND ROADS GEOJSON TO CAPTURE REAL-TIME DRIVER BEHAVIOR.

BY APPLYING A PRECISION LANE BUFFER (6-12M), WE CALCULATE THE AVERAGE SPEED SPECIFICALLY FORTHE ROAD SEGMENT WHERE THE SIGN IS LOCATED, FILTERING OUT NOISE FROM PARALLEL OR INTERSECTINGROADS.

STEP 4: DATA FUSION & SCORING

THE FINAL STAGE IS THE CROSS-SOURCE CORRELATION. WE CORRELATE THE VISION FINDINGS WITH THEBEHAVIORAL FINDINGS.

BEHAVIORAL VALIDATION

The Probe Layer

We integrate raw **HERE Probe Data** to capture real-time driver physics.

Precision Lane Buffer (6-12m)
We isolate the average speed specifically for the road segment where the sign is located, filtering noise from parallel roads.



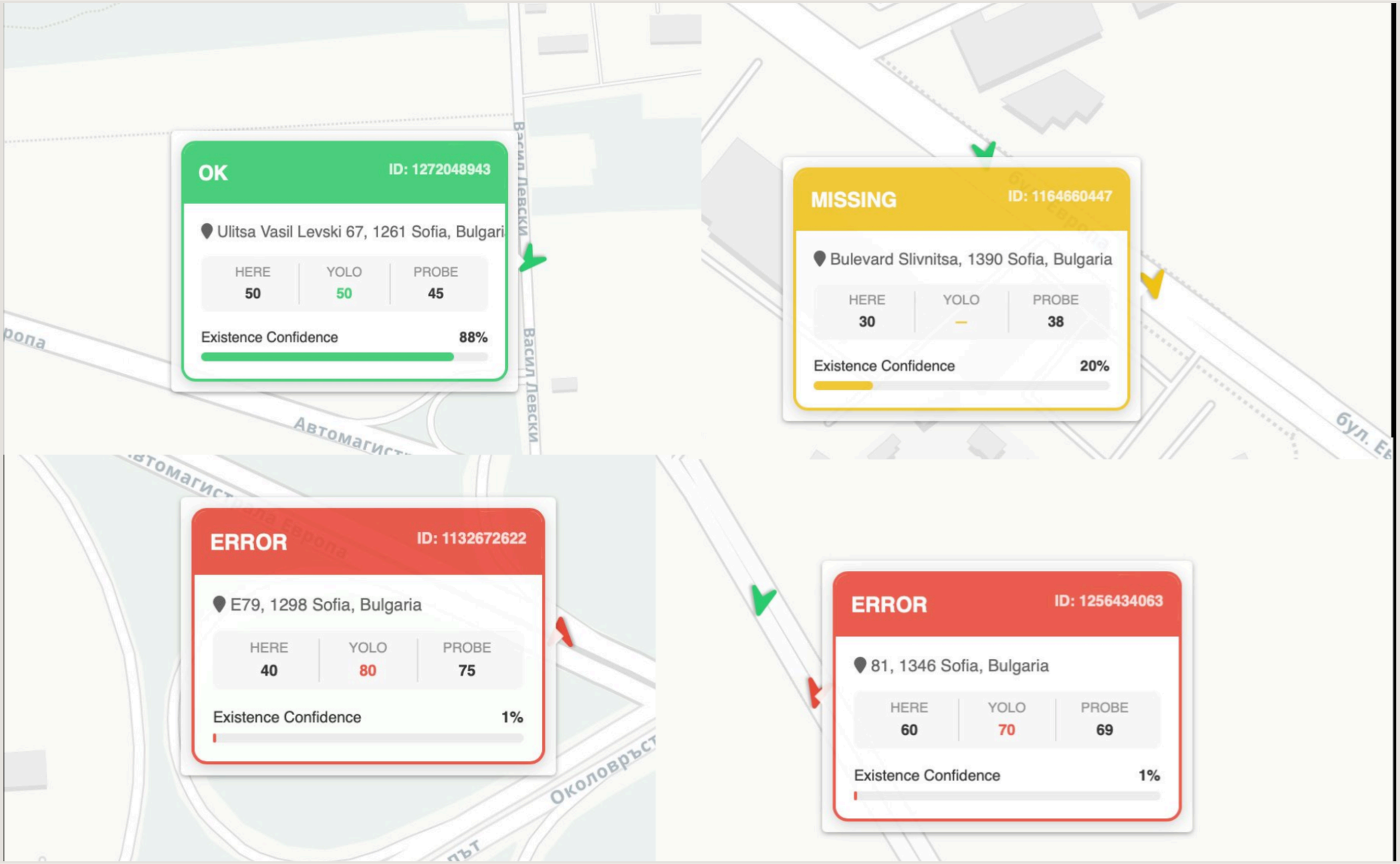
THE TRUTH METER

Existence Confidence Score

We calculate the final reliability of a map node by correlating Vision findings with Behavioral physics.

$$\text{ECS} = C_{\text{vis}} \cdot e^{-\frac{|V_{\text{det}} - S_{\text{prob}}|^2}{2\sigma^2}}$$

Our Gaussian similarity function penalizes the score as the gap between AI detection and Driver speed grows. This acts as a **Safety Fuse** for the map.

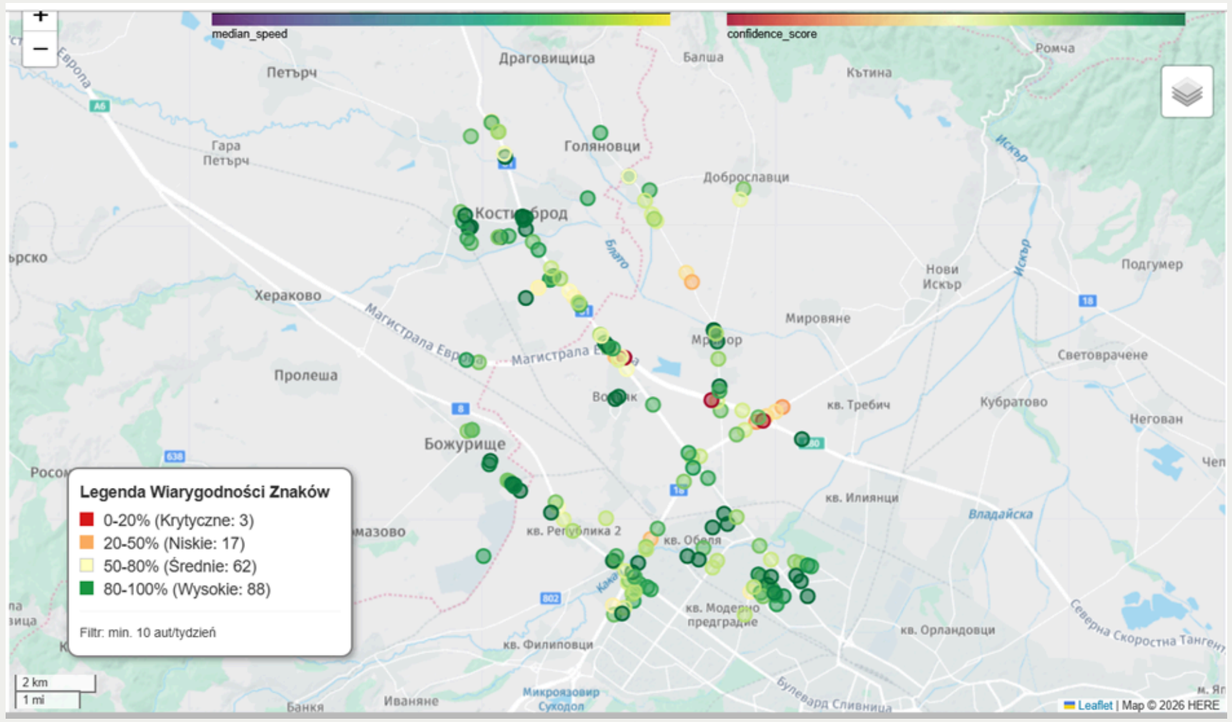

OK
ID: 1272048943
Ulitsa Vasil Levski 67, 1261 Sofia, Bulgari
HERE
50
YOLO
50
PROBE
45
Existence Confidence
88%
MISSING
ID: 1164660447
Bulevard Slivnitsa, 1390 Sofia, Bulgaria
HERE
30
YOLO
–
PROBE
38
Existence Confidence
20%
ERROR
ID: 1132672622
E79, 1298 Sofia, Bulgaria
HERE
40
YOLO
80
PROBE
75
Existence Confidence
1%
ERROR
ID: 1256434063
81, 1346 Sofia, Bulgaria
HERE
60
YOLO
70
PROBE
69
Existence Confidence
1%

The 30 vs 40 Scenario

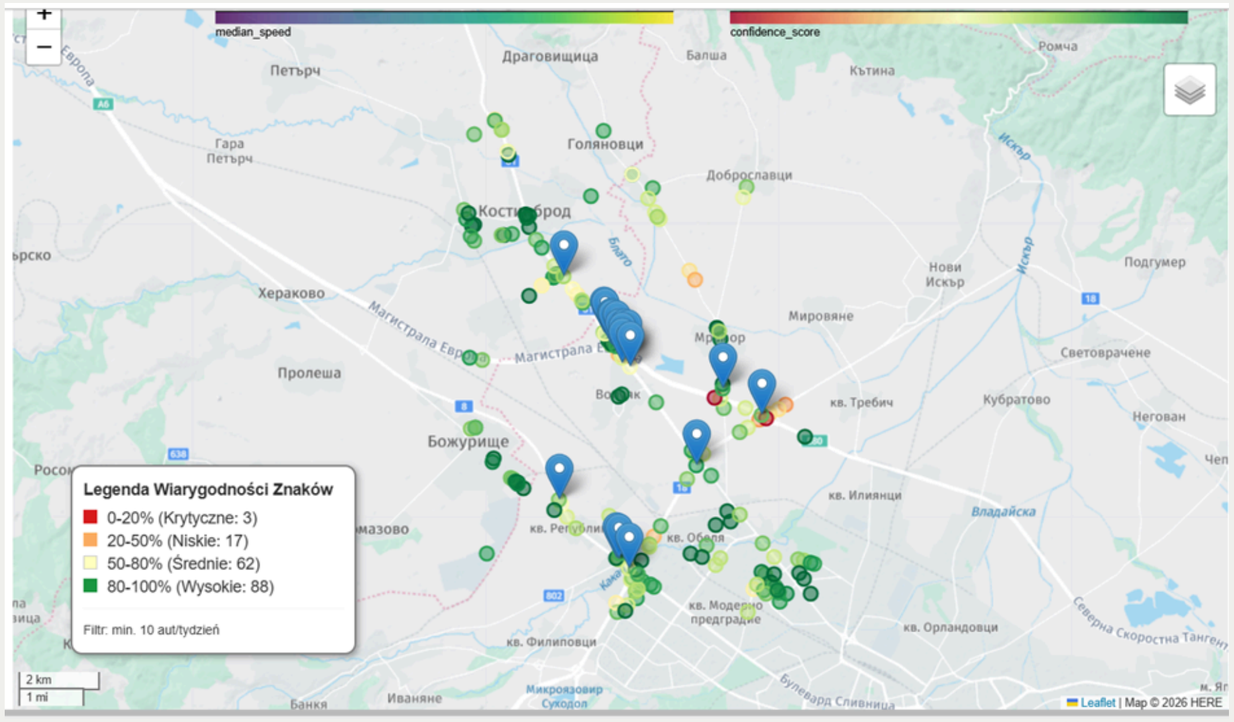
Scenario	Map Limit	AI Vision	Probe Speed	Final ECS
Outdated Map	30 km/h	40 km/h	38 km/h	0.88 (HIGH)
AI Error	30 km/h	100 km/h	32 km/h	0.05 (LOW)
Verified OK	50 km/h	50 km/h	48 km/h	0.95 (HIGH)

When AI and Drivers agree (Case 1), the map is proven wrong. When they conflict (Case 2), the system protects the map from AI errors.

VISUAL EXTRACTION



map shows points from PROBE DATA

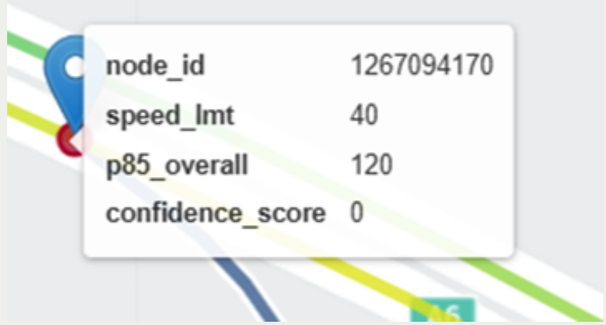


map shows points from PROBE DATA & photos analysis effect

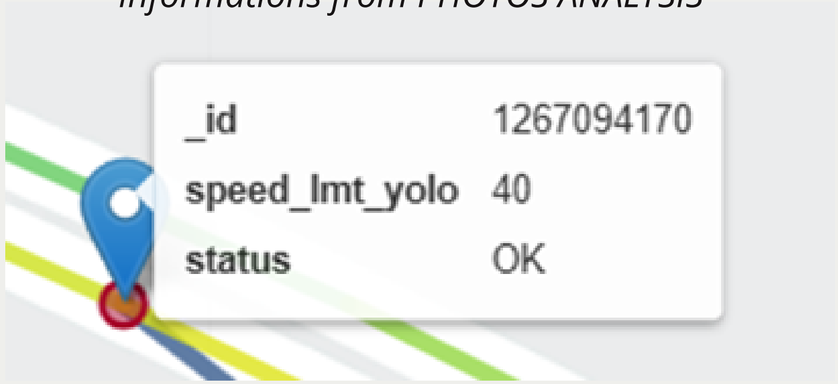
CRITICAL POINTS ANALYSIS



informations from PROBE DATA

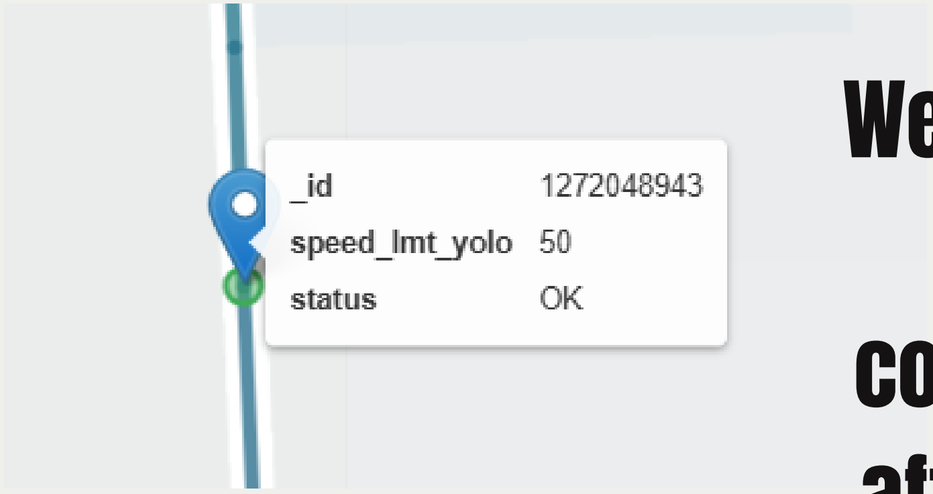


informations from PHOTOS ANALYSIS

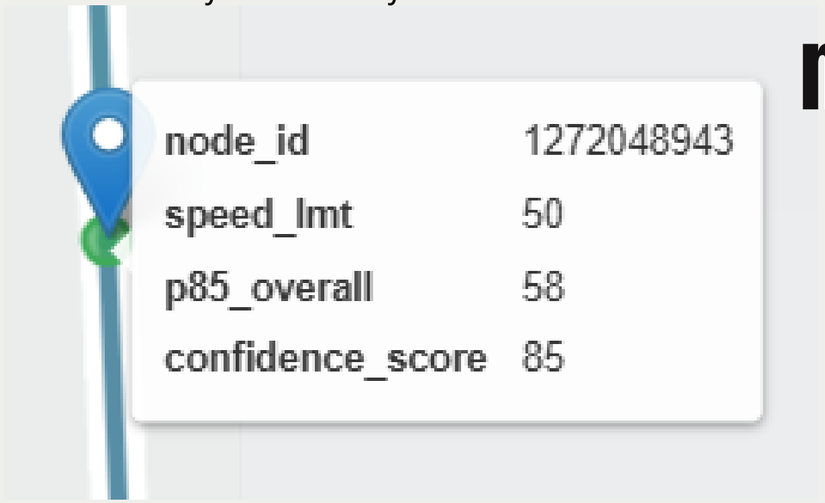


PERFECT ALIGNMENT: YOLO VS PROBE DATA

informations from PHOTOS ANALYSIS



informations from PROBE DATA



We cross-referenced the visual extraction with system probe data, including a comparison of speed medians before and after the sign. The results show a perfect match, confirming high data integrity and model reliability

Business Value & ROI

60%

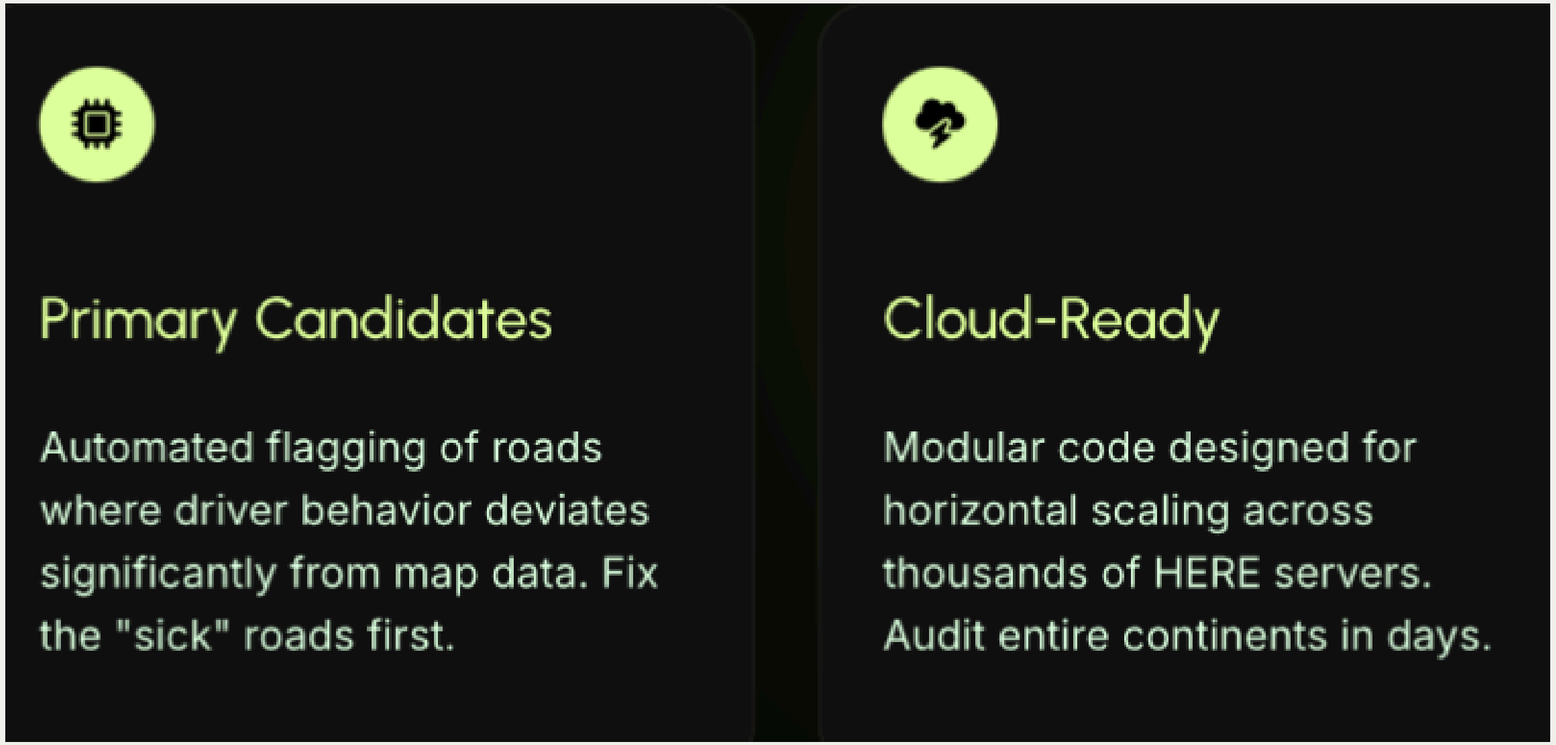
EFFICIENCY GAIN

Optimized Maintenance

Automate verification of "OK" spots. Editors focus 100% of their energy only on critical errors detected by our system.

100% ISA Compliance: Ensure HERE provides the most accurate speed limits, reducing liability and increasing safety.

BUILT FOR SCALE



THANK YOU

GEOLOGIC

Kamila Bodziony

Aleksandra Staroch

Mateusz Grodzki